



Saudi Arabian Mining Company (MAADEN)

Digitalization Scope of Work (SoW)

Project: Development of an Integrated, AI-Powered Digitalization Platform for Maaden

ARGP-DIGI-APP-SOW-001

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1.Introduction

1.1 Project Overview

The Ar Rjum Gold Project (ARGP) is a flagship greenfield mining and processing development for Maaden in the Kingdom of Saudi Arabia. The project's scale and complexity necessitate a robust, AI-enabled digitalization platform to ensure schedule certainty, capital efficiency, and operational excellence from design through to startup.

1.2 Strategic Importance

ARGP is a cornerstone of Maaden's growth strategy and a key contributor to Saudi Vision 2030. This digital platform will enhance local content, strengthen supply chain resilience, and embed transformative technologies, establishing a replicable blueprint for future mega-projects.

1.3 Scope Framing Assumptions & References

This SoW mandates adherence to Maaden's internal standards (MCIS, Engineering), international benchmarks (ISO, IEC, CII), and strict compliance with KSA cybersecurity (NCA), data privacy (PDPL), and cloud computing (CST) regulations.

1.4 Key Risks & Mitigations

- **Data Quality:** Implement automated data cleansing and scoring.
- **User Adoption:** Establish power-user champions and comprehensive training.
- **Cybersecurity:** Achieve early alignment with NCA/CST; prefer in-Kingdom hosting.
- **Vendor Lock-in:** Enforce open APIs and document clear exit strategies.

2.Overall Platform Vision & Strategic Goals

The proposed platform must serve as the single source of truth for the ARGP, achieving the following strategic goals:

- **Unified Data Environment:** Centralize all project data, breaking down silos between all stakeholders.
- **Predictive Project Controls:** Utilize AI/ML for proactive insights into schedule, cost, quality, and safety risks.
- **Digital Workforce Transformation:** Enable efficiency through digital work packaging, constraint management, and real-time progress validation.
- **Seamless Digital Handover:** Ensure a flawless, data-rich transition from construction to commissioning and operations.
- **Intelligent Compliance & Assurance:** Automate enforcement of standards and provide data-driven quality and performance assurance.

2.1. Scope Framing Assumptions & References

This Scope of Work (SoW) assumes adherence to Maaden's MCIS compliance, Maaden Engineering standards, international benchmarks such as ISO, IEC, and CII frameworks, and strict compliance with the Kingdom's cybersecurity, data residency, and regulatory frameworks. It is based on the principle of leveraging digital transformation, AI integration, and predictive analytics to enhance design efficiency, safety, and constructability across the project life cycle.

2.2. Risks & Mitigations

Key risks and mitigations include:

- Data quality risks → Implement cleansing and scoring mechanisms.
- Adoption resistance → Establish engineering champions and training workshops.
- Cybersecurity approvals → Early alignment with NCA and CST authorities. Vendor lock-in → Enforce open APIs and provide clear exit strategies.

3.Detailed Scope of Work & Functional Modules

3.1: Scope of Work – Engineering & Design Management

3.1.1: Objectives

The objectives of this SoW are multi-dimensional, focusing on delivering engineering excellence through digital transformation and AI-driven solutions. Key objectives include

- Establishing a Common Data Environment (CDE) that ensures all engineering deliverables are centralized, traceable, and governed by Maaden's standards.
- Introducing AI-driven optimization tools for compliance checks and detecting design clashes, a demo is required to show system capability and enhance overall assurance.
- Building a project-wide digital twin to simulate constructability, safety, and operability before execution.
- Embedding predictive analytics to provide foresight into overall project bottlenecks, and potential rework scenarios with respect to all aspects.
- Driving knowledge management by codifying lessons learned into AI-driven engineering rule libraries.

3.1.2: Functional Requirements for Software Developer Integration

Software Developer 's platform must serve as the backbone of ARGP's digital engineering ecosystem, providing the following functional capabilities but not limited to listed hereunder,

1. Centralized data environment for version-controlled engineering deliverables.
2. Revit/Navisworks, Plant 3D, P&ID tools, federated models for clash avoidance and design validation.
3. Digital twin platform or engineering data backbone, middleware to connect 3D Model, P&ID, Engineering Documents, Engineering indexes for all disciplines and field data
4. Philosophy-based data model mapping units, systems, subsystems, and tags to ensure traceability.
5. Design layer: 3D models, P&IDs, equipment specs, loop diagrams Real-time change impact analytics to quantify downstream effects on cost, schedule, and constructability, also Risk modeling for scope changes, delays and third-party issues.
6. Asset and data layer: asset catalog, equipment tags, instrument loops, utilities, cables.
7. Workflows layer, checklists, engineering procedures, approval workflows, non-conformance management for design phase.
8. AI-powered reporting dashboards covering engineering maturity, RFI turnaround, model review performance, and schedule predictability.

3.1.3: Engineering Data Model & Integrations

The engineering data model shall be Semantic model-driven, structured around a hierarchy of Areas → Units → Systems → Subsystems → Tags → Documents. This framework will ensure consistency across design packages, procurement activities, and construction sequencing. Integrations will be API-based, enabling seamless connectivity between design tools, vendor data repositories, construction planning platforms, and quality inspection systems with AI driven and optimization.

3.1.4: Engineering Data Management & Security

The engineering data environment must comply with ISO/IEC 27001 Information Security Management standards, while aligning with Saudi NCA Essential Cybersecurity Controls (ECC-1:2018), PDPL, and CST Cloud Computing Regulations. Data residency requirements mandate that all project data remain within KSA. Strict role-based access controls, immutable audit trails, vulnerability assessments, and disaster recovery protocols must be embedded to safeguard data integrity and system resilience.

3.1.5: AI-Enabled Design & Constructability Checks

AI will be deployed to automate compliance validation against Maaden standards, detect design clashes, and evaluate constructability at early design stages. Software Developer should utilize the best approach, and tools to analyze technical specifications, vendor documents, and RFIs to ensure alignment with design intent. Predictive clash detection and logistics optimization algorithms will minimize costly rework and enable proactive resolution of constructability risks.

3.1.6: Digital Twin & Predictive Engineering Optimization

The project will deploy a comprehensive digital twin integrating engineering, procurement, and construction data.

This digital twin will enable scenario simulation (e.g., tie-in sequences, modularization feasibility, interface points validation, interdisciplinary validations etc.) and predictive monitoring of critical equipment and systems. By benchmarking against global best practices, the digital twin will ensure continuous design optimization, risk minimization, and operational readiness.

3.1.7: Deliverables, KPIs & Acceptance Criteria

Deliverables include:

- Fully functional CDE with integrated Software Developer platform.
- AI compliance engine with automated pass/fail workflows.
- Project digital twin for Interdisciplinary verification, Interfaces clashes, Engineering mismatches, incomplete tag assignment, missing engineering data link etc.
- Dashboards track design maturity, Engineering phase schedule risks, interfaces, Deliverables documents etc.
- Further deliverable parts of the module but not limited to below,
 - Show case the utilization of set standard in equipment design and check their compliance.
 - Ability of performing equipment design calculations.
 - Design alternative Scenarios testing and identifying optimum design option with comparison.
 - Navigating through our engineering standards and project specifications as an AI solution.
 - Provide technical design resolution with references to standards and specifications.
 - Reflect our previous design lesson learnt as inputs to the design model of equipment.
 - Plot Plan optimization for lower bulk material considering access points, maintenance areas, construction lay down areas etc.

- Quantify benefits in terms of cost, schedule, quality etc. of using the solution.
- Show the ability of predicting design flaws in future.
- Capability of dealing with design assumptions and holds and suggest countermeasures.
- Ability of design package review rather than single equipment if applicable.
- Provide design completion percentage for equipment, packages and overall design including percentage of compliance to project standards and specification. In addition, ability to analyze and navigate through the data.

KPIs include:

- ≥30% reduction in RFIs prior to IFC release.
- ≥25% reduction in design rework hours.
- ≥20% improvement in design calculation and validations.
- ≥15% increase in engineering constructability alignment during model reviews.

3.1.8: Roles, Responsibilities & RACI

Responsibilities are distributed as follows:

- Software Developer : Configure platform, deploy AI solutions, ensure compliance, and deliver training.
- Maaden Engineering & Innovation: Define standards, approve rule libraries, and validate deliverables along with Project Teams members.

3.1.9: Implementation Plan & Timeline (Pilot → Scale)

The implementation will be executed in phases:

- Weeks 1–2: Mobilization, Philosophy setup, compliance framework mapping.
- Weeks 3–4: Initial integration pilots with CAD and procurement platforms.
- Weeks 5–6: Deployment of AI rule libraries, clash detection, and constructability analytics.
- Weeks 7–8: Validation of pilot deliverables, KPI tracking, and planning for full-scale rollout.

3.1.10: Discipline-Specific Design Checks and Calculations

A. Architectural

- Validate building codes compliance (IBC/IFC), life-safety egress, occupancy, fire-rating of walls/doors, blast/overpressure where applicable.
- Space planning checks against equipment footprints, clearances, maintenance envelopes, and lifting paths.
- Material suitability for corrosive/abrasive environments; floor loading vs. equipment and live loads.
- Building envelope performance: environmental sealing, thermal, acoustic, and dust ingress protection per project climate.
- Accessibility/ADA as applicable; slope/drainage checks; coordination with HVAC, electrical, fire systems.

B. Mechanical Equipment (Static and Rotating)

- **Static equipment:**
 - Material selection against process fluid composition, temperature/pressure, corrosion/erosion; internal linings; corrosion allowance.
 - Thickness/calcs per ASME VIII/API 650/620; nozzle loads per WRC; saddle/skirt design; seismic/wind loads; NDT/inspection class.
 - Design pressure/temperature margin, MDMT; relief device sizing per API 520/521; nameplate checks vs. datasheets.
- **Rotating equipment (pumps, compressors, mixers, ball mills, etc.):**
 - Hydraulic/thermodynamic sizing vs. duty points; NPSH margin; efficiency and operating range; API compliance (e.g., API 610/617).
 - Driver sizing (motor/turbine): power/torque, service factor, enclosure, area classification, ambient; starting method; VFD compatibility.
 - Coupling, alignment, vibration criteria, baseplate/grouting; seal/flush plans; lube systems.
- **Operability/reliability AI checks:**
 - Duty/standby logic, sparing, run out/stonewall risks, surge control feasibility; maintenance access; historical anomaly patterns.

C. Isometrics

- Dimensional consistency: line numbers, sizes, specs, ratings vs. line list and P&IDs.
- Slope and high-point/low-point vents/drains for cleanability and air/pocket elimination.
- Weld/joint types, shop vs. field weld splits; field weld factor control.
- Support tags and locations matching stress model; accessibility for NDE; prefabrication feasibility.
- Material take-off reconciliation with 3D and P&IDs; hydrotest limits and test medium notes.

D. Master Equipment List (MEL)

- Tag uniqueness and hierarchy (Area-System-Subsystem); service descriptions; design vs. operating conditions.
- Datasheet completeness scoring; vendor data integration; spare philosophy alignment.
- Inter-discipline linkages: electrical loads, instrument I/O, foundations, lifting devices.

E. Piping Special Items

- Flex hoses, expansion joints, strainers, orifice plates, spectacle blinds, injection quills, sampling points:

- Material compatibility, rating, face-to-face, orientation; correct placement per process needs.
- Expansion joint movements, anchors/guides; strainer mesh selection and differential pressure limits.
- Positive isolation provisions and double-block-and-bleed where required.

F. Structure

- Steel and concrete design checks per AISC/ACI; load combinations (dead, live, wind, seismic, thermal, equipment dynamic).
- Pedestals, foundations, anchor bolts, grout pocket sizing; vibration and deflection limits for rotating equipment.
- Corrosion protection, fireproofing, coating systems; platform/grating loading and stair/ladder compliance.
- Lifting beams/monorails: capacity, deflection, headroom; equipment maintenance clearances.

G. Electrical Equipment

- Equipment suitability: IP/NEMA rating, ambient, altitude, derating; area classification (IECEX/ATEX).
- LV/MV/HV switchgear: short-circuit withstand, bus ratings, arc-flash boundaries; protection relay settings envelope.
- UPS and DC systems capacity; battery autonomy; charger sizing and redundancy.
- Motor control centers: starter type/VFD harmonics; thermal limits; load lists vs. feeder sizing.

H. Electrical (Power Systems)

- Load flow, short-circuit (IEC 60909/IEEE), motor starting, voltage drop, earthing, touch/step voltage; transformer sizing and impedance.
- Cable sizing: ampacity, derating (ambient, grouping, soil), voltage drop, short-circuit thermal withstand; XLPE/EPR selection.
- Cable routing: 3D-detected lengths and routes; tray fill including minimum 20% spare; segregation (power/control/FO); EMC.
- Lighting design: lux levels, uniformity, task/emergency lighting, hazardous area suitability, glare, energy metrics.

I. Instrumentation

- Instrument index consistency; device selection vs. process range, turndown, accuracy, wetted materials; IP/temperature class.

- Installation requirements: tapping locations, straight-run lengths, impulse line heat tracing; filter/condensation arrangements.
- Control valves and ON/OFF valves: Cv sizing, pressure drop, noise, cavitation/flashing, actuator sizing, fail position, SIL capability.
- Multipair cable planning: signal allocation balancing across cables to avoid single-point failures; spare pairs management.
- Junction box location optimization: minimize home-run lengths; environmental/IP rating; segregation of IS/non-IS circuits.

J. Control System

- DCS/PLC architecture: controller loading, network topology, redundancy, time sync; cybersecurity zones and conduits (IEC 62443).
- I/O assignment and marshalling checks; HART/Fieldbus/Profinet compatibility; intrinsic safety entity/loop calculations.
- Cause & effect, alarm philosophy conformance, alarm load analysis; interlock and trip validation vs. SIF/SIL targets (IEC 61511).
- Graphics and naming consistency; loop integrity and loop tuning readiness checks.

K. Cables

- Type and construction verification (armoring, shielding, insulation, fire performance); ultraviolet/chemical resistance.
- Voltage drop and short-circuit checks for all circuits (UPS/non-UPS, single/three-phase, MV/HV).
- Instrumentation cable noise immunity; pair/twisted shield requirements; maximum loop resistance and capacitance.
- Earthing and bonding continuity; gland/termination compatibility; pulling tension/bend radius constraints.

L. Test Pack

- Systemization and sub-systemization mapping; hydro/pneumatic test boundaries reconciliation with P&IDs and isos.
- Test pressure and medium checks per code; reinstatement items; punch list categorization and turnover readiness metrics.

- Weld/NDE traceability; calibration certificates linkage for instruments in test loops.

M. Valves

- Pressure class, end connections, materials (body/trim), temperature ratings; fugitive emissions compliance.
- Fire-safe, antistatic, low-temperature service as applicable.
- Actuated valves: torque margins, speed, supply pressure/voltage; partial-stroke test capability; position feedback.
- Severe service trims and noise/erosion checks; seat leakage class; PM plan recommendations.

N. Civil

- Geotechnical data integration; foundation bearing checks; settlement and differential movement limits.
- Concrete mix exposure class, cover, crack control; chemical attack resistance per process environment.
- Drainage, paving, roads; spill containment and slope verification; underground services clash checks.
- Buildings and pits water ingress protection; cathodic protection interfaces for buried metallics.

O. Safety

- Process safety compliance: relief, blowdown, flare capacity tie-in checks; hazardous area classification and equipment conformity.
- Escape route analysis; muster and access; safety signage and eyewash/shower locations.
- Fireproofing extent and thickness; structural response in fire; hazardous inventory isolation logic.
- Quantitative risk insights using AI anomaly detection for high-risk clusters and mitigations.

P. HVAC

- Heat load calculations; ventilation rates; pressurization schemes; hazardous area ventilation and purge checks.
- Filtration class, corrosion protection for coastal/desert environments; duct routing clashes; noise and vibration limits.
- Package unit selections vs. ambient extremes; redundancy and control strategy validation.

Q. Fire and Gas

- Detection coverage analysis (gas, flame, smoke, heat); spacing and height; voting logic and response times.
- Firewater demand and hydraulics; hydrants/monitors coverage; sprinkler/deluge design density.
- F&G cause and effect matrix validation; integration with ESD/DCS; hazardous zoning alignment.

R. Spools

- Spool break logic for fabrication and transport; weld map and numbering; NDE extent.
- Fit-up allowances; field weld accessibility; paint/lining sequence; pick-up points and weights.

S. System and Sub-System

- Definitive systemization tree; System, Sub-System, Engineering package handover boundaries, Engineering punch list classification and closure workflow.
- Engineering base system identification with simulation modules to identify system, sub-system limitations and interface points.
- Progress metrics at system/subsystem level, System Boundaries adjustment, etc.

T. Cross-Discipline Consistency Checks

- P&IDs vs. 3D model vs. isometrics vs. line list vs. valve list consistency; AI-based delta detection and reconciliation suggestions.
- Equipment datasheets vs. vendor GA vs. foundations vs. electrical loads vs. cable schedules.
- Hazardous area classification vs. equipment/cables/instruments certifications and installation details.
- Process conditions vs. material selection across equipment, piping, valves, instruments.
- Control narratives vs. logic diagrams vs. I/O allocations vs. cause & effect.
- Cable routing vs. tray capacities vs. segregation rules; ensure 20% spare maintained.

3.2. Scope of Work – Schedule Management & Project Controls

3.2.1: Objectives

- Stand up a single source of schedule truth with robust governance and auditable change control.
- Apply AI-assisted analytics to diagnose model health, forecast outcomes, and recommend prioritized mitigations.
- Critical path analysis for schedule optimizations
- Optimize schedule duration by 10%
- Evaluation and Mitigation of Key Project Risks
- AI Based Comparison of Installation Rates for Key Construction disciplines across the Globe with GCC region.
- AI based validation of the Baseline schedule by generating various scenarios for schedule optimization.
- Key Construction quantity plan analysis based on region wise AI database
- Execute quantitative schedule risk analysis (QSRA) to set realistic confidence dates and time contingency, by conducting workshop involving all stakeholders for obtaining the optimistic & Pessimistic durations for the critical activities.
- Operationalize AWP-aligned look-ahead and constraint removal with measurable plan-percent-complete (PPC) gains.
- Integrate cost-schedule performance (EVMS) to separate productivity issues from plan quality issues.

3.2.2: Scope Boundary (Pilot → Scale)

- The pilot will target high-impact areas (e.g., process plant civils/structural, utilities interfaces, logistics corridors) to demonstrate measurable schedule risk reduction and decision velocity.
- Upon KPI attainment, the solution scales across remaining work packages with standard connectors, governance rituals, and reusable analytics patterns.

3.2.3: Software Developer Integration & Data Sources

The platform shall ingest, model, and relate multi-source data with full provenance and automated refresh:

- Primavera P6: baseline/current schedules, calendars, WBS, codes, and logic (XML/API).
- Risk Register: qualitative/quantitative risks, threat/opportunity flags, owners, and mitigation plans.
- Procurement: requisitions, POs, expediting, logistics milestones, material availability.
- Construction: field quantities, time sheets, progress claims, QA/QC inspections, punch lists.
- Permits/Interfaces: utilities, right-of-way, third-party approvals, tie-in windows.
- Change: BEI/TIA analyses, RFC/FCN/TCN, trends and variance attribution.

3.2.4: Schedule Governance & Model Quality

Govern schedule model standards, calendars, and coding structures. Enforce a cadence of baseline approval, updates, variance analysis, and change control to maintain credibility and dependability of dates.

- Baseline integrity: versioning with diffs on logic, constraints, and calendars; audit trail for approvals.
- Coding standards: WBS, areas/units/systems, CWP/IWPs, cost accounts, interfaces; resource codes if applicable.
- Quality gates: duration reasonableness, float distribution, logic density, limited artificial constraints.
- AI based comparison of planned durations of Baseline with Industry standards for validation of the durations.
- Manpower mobilization analysis with respect to productivity rates of GCC region. Quantity plan analysis based on benchmark installation rates.

3.2.5: Schedule Health Diagnostics (DCMA/GAO)

Automate industry-accepted checks (DCMA 14-Point; GAO best practices). Render exception lists with heatmaps and drill-downs by WBS, area, and contractor.

- DCMA metrics (e.g., missing logic, leads/lags, constraints, invalid dates, negative float, high duration).
- GAO quality pillars: comprehensive, well-constructed, credible, and controlled; checklists and trends embedded.
- Recurring violation alerts with owners, due dates, and SLA tracking.
- Recommendations for schedule health improvements

3.2.6: Quantitative Schedule Risk Analysis (QSRA)

Perform Monte Carlo simulations on the CPM model with risk-driver distributions and correlation controls. Output confidence curves (P-dates), driver tornado charts, and quantified mitigation scenarios with expected time recovery.

- Three-point ranges at activity groups; logic uncertainty modeling; risk calendars and correlation matrices.
- Mitigation portfolios ranked by time gain, feasibility, cost, and HSE implications.
- Contingency governance: time contingency sizing, drawdown rules, and decision thresholds.

3.2.7: What-If Simulation & Optimization

Model resequencing, alternate methods, resource strategies, and interface timing options. AI support highlights the smallest set of changes with the largest impact on critical/near-critical paths while preserving realism.

- Constraint-aware optimization: access, materials, permits, utilities; readiness scoring drives plan inclusion.
- Shift patterns and calendar experiments; fatigue, HSE, and constructability constraints encoded.
- Utility energization and corridor handover scenarios with ripple-effect visualization.
- AI based solution like conversion of Stick built to modularization of CWP for optimization of the schedule.
- Identification of critical activities which can be performed in parallel to optimize the schedule.
- Generation of construction scenarios based on the Manpower mobilization.
- AI based acceleration schedules to bring back the delayed schedule on track.
- Generation of scenarios based on Low productivity and trend analysis.

- Optimization of critical Equipment erection sequence based on the AI inputs (database)
- Manpower density analysis by integration with 3D model of the CWP

3.2.8: Integrated Cost-Schedule Performance (EVMS)

Fuse time and cost to compute SPI/CPI, SV/CV, TCPI, EAC/ETC, and trends. Separate genuine productivity issues from schedule model weaknesses and quantify recovery.

- Time-phased baselines tied to work packages; rules of credit consistent with field measurement.
- Contractor performance scorecards: schedule health, EV indices, change churn, claims exposure.
- Surveillance pack aligned to credible EV practices; corrective action logs with root-cause taxonomy.
- Forecast EAC based on the schedule analysis.

3.2.9: AWP Alignment & Look-Ahead Management

Align to the Path of Construction (EWP→CWP→IWP). Only 'ready' work advances into the 2–6 week look-ahead. Visual management drives workforce productivity and reduces standby.

- Digital CWPs/IWPs with readiness: design, materials, access, permits, tools, and pre-requisites.
- Workforce heatmaps: constraint type, age, and criticality; PPC uplift tracked weekly.
- Rapid feedback loop from field to planning to de-bottleneck.

3.2.10: Interface, Constraint & Permit Management

Treat interfaces and constraints as first-class schedule objects with owners and SLAs. Automated aging reports drive accountability and timely removals.

3.2.11: Progress Measurement & Rules of Credit

Define discipline-specific rules of credit (RoC) for quantity-based progress and align with payment mechanisms. Reconcile claimed vs. measured progress using field evidence.

- Standard RoC libraries by discipline; tie to cost accounts and pay items.
- Except workflows for anomalies between claimed and verified quantities.
- Automated earned progress roll-up at WBS/area/contractor levels.
- Integration of Progress validation by discipline engineers with Quality (Inspection Reports)
- Automated Verification of claimed quantity from the IFC drawings of CWP.

3.2.12: Procurement & Logistics Synchronization

Integrate material readiness and logistics with schedule promises. Prevent start-stop cycles by ensuring materials, tools, and vendor support are synchronized with Workfront plans.

- Need-by vs. on-site curves; expediting alerts tied to critical work fronts.
- Laydown and handling constraints incorporated into look-ahead feasibility.
- Vendor interventions (inspections, FAT, SAT) embedded in logic with owners.
- Automated integration of Procurement status with Control schedule for generating schedule alerts.

3.2.13: Commissioning, Start-Up & OR Readiness Linkage

Link construction completion to subsystem pre-commissioning, commissioning, and RFSU. Surface blockers and punch aging; monitor readiness KPIs and turnover documentation maturity.

3.2.14: Executive Dashboards & Decision Support

Provide tiered dashboards: portfolio (executive), project (controls), and area/package (construction). Summarize risk exposure, milestone confidence, and the top five recovery actions each cycle

3.2.15: Data Security, Residency & Compliance (KSA)

- ISO/IEC 27001-aligned ISMS; least-privilege access, MFA, encryption at rest/in transit, immutable logs.
- NCA Essential Cybersecurity Controls (ECC-1:2018); KSA PDPL; CST Cloud Computing Regulatory Framework and Cloud-First policy; preference for in-Kingdom hosting.
- Third-party/PMC segregation, EULA, update cadence, server location disclosure, DR/BCP testing.

3.2.16: Deliverables, KPIs & Acceptance Criteria

Deliverables

- D1. Software Developer –P6 integration with governance workflows and lineage.
- D2. Automated DCMA/GAO health dashboards and exception management.
- D3. QSRA engine and reports (S-curves, tornado drivers, mitigation quantification).
- D4. What-if/optimization scenarios and decision logs.
- D5. EVMS integration and contractor scorecards.
- D6. AWP-aligned look-ahead with constraint Kanban and readiness metrics.
- D7. Security & compliance dossier (PDPL/ECC/CCRF), runbook, and training.

Pilot KPIs (Targets)

- K1. ≥20% reduction in high-risk activities on critical/near-critical paths within two reporting cycles.
- K2. ≥15% improvement in look-ahead PPC and constraint clearance.
- K3. ≥10–15% SPI trend improvement on pilot packages without artificial logic manipulation.
- K4. QSRA confidence uplift ≥15 percentage points for the next major milestone.
- K5. ≤5% variance between claimed vs. measured progress on two critical CWP.

Acceptance Criteria

- All deliverables D1–D7 implemented and demonstrated; user acceptance by Project Controls and Construction.
- KPIs K1–K5 achieved on pilot scope or justified with evidence and improvement trajectory.
- Security/compliance artefacts approved by IT/Cybersecurity; end-user training completed.

3.2.17: Roles, Responsibilities & RACI

- Vendor (Software Developer): Configure integrations, diagnostics, QSRA, optimization, dashboards; compliance mappings; training.
- Owner Project Controls: Baseline ownership, updates, health gates, QSRA inputs, mitigation governance.
- Construction Management: Field progress, constraints, readiness; execute recovery actions and feedback.
- Procurement: Material status, expediting, logistics data; align need-by with plan.
- IT & Cybersecurity: Hosting validation, residency compliance, audits, DR/BCP.
- Document Control: Change logs, approvals, and audit trails.

3.2.18: Implementation Plan & Timeline

Weeks 1–2: Mobilization, data inventory, P6 connector, governance standards, security design.

Weeks 3–4: Health checks live; initial QSRA; mitigation playbook baseline; dashboard pilots.

Weeks 5–6: Optimization scenarios; EVMS integration; AWP look-ahead with constraint Kanban.

Weeks 7–8: Adoption clinics; KPI measurement; executive read-out; scale plan and budget.

3.2.19: Risks, Assumptions & Dependencies

- Data quality and logic integrity → institute schedule quality gates and fix-forward sprints.
- Adoption resistance → leadership sponsorship, coaching, and visible wins.
- Cyber approvals timeline → early submissions; KSA-resident hosting preference.
- Contractor participation → contract language for data timeliness and quality; scorecards tied to pay.

3.2.20: Standards & Best-Practice References

This SoW aligns with leading practices for megaproject schedule & controls, including but not limited to:

- PMI – Practice Standard for Scheduling (3rd Ed.).
- GAO Schedule Assessment Guide – Ten Best Practices for Developing High-Quality Schedules.
- DCMA 14-Point Assessment; DCMA DECM surveillance context for EVMS credibility.
- AACE Recommended Practices: 57R-09 (Integrated Cost & Schedule Risk Analysis), 64R-11 (CPM Schedule Risk Modeling), 49R-06 (Identifying the Critical Path), 29R-03 (Forensic Schedule Analysis).
- ISO 21502 – Project Management Guidance; ISO 31000 – Risk Management Principles and Guidelines.
- CII – Advanced Work Packaging (AWP) / Path of Construction principles.

3.2.21: Vendor Response Requirements (Software Developer)

- Demonstrate robust P6 integration, governance workflows, and lineage preservation.
- Run a QSRA on anonymized sample data; present S-curves, tornado drivers, and quantified mitigations.
- Show what-if/optimization capabilities with constraint awareness and impact tracking.
- Provide EVMS integration, contractor scorecards, and executive dashboards with drill-downs.
- Submit PDPL/ECC/CCRF compliance mappings; KSA residency options; EULA, update cadence, and support SLAs.
- Provide training, runbook, and scale-out plan with resource model and timelines.
- Provide requirements related to Baseline schedule, such as Level 4, Resource loaded (Quantity & Craft wise Manpower) Cost loaded for conducting schedule & Cost Optimization Requirements related to Cost for forecasting EAC
- Implementation Plan for setting up the Construction Management Software

Additional Explicit Requirements

The solution must explicitly support the following capabilities, as emphasized by the Vice President in the meeting summary and subsequent email thread:

- End-to-end schedule data build: ingest, create, and structure schedule networks (activities, logic, calendars, codes) directly in the platform where needed, with full lineage and audit trail.
- Simulation at scale: run large-scale scenario simulations on the CPM model, including Monte Carlo and deterministic 'what-if' cases, without practical limitations on the number of activities analyzed.
- Constraint-based optimization: use a constraint approach (materials, access, permits, utilities, vendor support, crew limits) to identify the optimal resequencing/strategy that compresses critical and near-critical paths while preserving realism and HSE constraints.
- Distribution analytics: compute and visualize distributions for key dates (P-dates), float erosion, activity/chain durations, resource loading, and path criticality indices; provide confidence intervals and sensitivity to risk drivers.
- Requirements-to-outcomes automation: allow users to declare desired outcomes (e.g., target RFSU date, maximum overtime budget, access window limits), automatically propose and score scenarios that best satisfy the requirements, and publish the recommended plan with a decision log.
- EAC configuration and analysis: integrate cost and schedule to configure and analyze Estimate at Completion (EAC) and ETC scenarios; display EAC distributions with drivers (productivity, rate variance, change churn) and link to schedule risk exposure to maintain a coherent time–cost forecast.

3.3. Scope of Work – Construction and Pre-Commissioning

3.3.1. Objectives

This Scope of Work aims to establish a unified, data-driven, and AI-enabled management framework for the Construction and Pre-Commissioning phases of the Maaden project specially of the Ar Rjum Gold Project (ARGP). The primary objectives are to:

Ensure construction execution is perfectly synchronized with engineering deliverables and the project master schedule.

Transform the construction workforce through Advanced Work Packaging (AWP), constraint removal, and real-time progress analytics.

Seamlessly transition constructed systems to Pre-Commissioning, ensuring mechanical completion and operational readiness are achieved efficiently and safely.

Leverage the Software Developer platform as the single source of truth for construction and pre-commissioning data, enabling predictive insights, proactive risk mitigation, and quantifiable performance improvements.

3.3.2: Project Objectives & Strategic Goals

The platform must be designed to achieve the following strategic objectives:

- **Unified Data Environment:** Create a single, centralized, and version-controlled data repository for all project information, breaking down information silos between Engineering, Construction, and Commissioning teams.
- **Predictive Project Controls:** Utilize AI and machine learning to provide predictive insights into schedule delays, cost overruns, quality issues, and safety risks, enabling proactive mitigation.
- **Workforce Digitalization:** Transform on-site execution through digital Advanced Work Packaging (AWP), constraint management, and real-time progress validation.
- **Seamless Handover:** Ensure a flawless and data-rich transition from Construction to Pre-Commissioning and from Pre-Commissioning to Commissioning and Operations, with all data and certifications digitally compiled and linked.
- **Contractual Compliance & Enforcement:** Provide the digital framework for contractors to meet all their contractual obligations, with automated tracking of approvals, tests, and deliverables.

3.3.3: Scope & Platform Architecture

Platform Core Architecture

- **Cloud-Native & Scalable:** A SaaS-based, multi-tenant architecture capable of scaling to support multiple mega-projects simultaneously.

- **Open API Framework:** Robust, secure APIs for bidirectional integration with all core project systems (e.g., ERP, CAD/BIM, Scheduling (P6), Document Management, IIoT).
- **KSA Data Residency:** All data must reside within the Kingdom of Saudi Arabia, compliant with NCA ECC-1:2018, PDPL, and CST Cloud Computing Regulatory Framework.
- **Semantic Data Model:** A flexible, ontology-driven data model that can map and relate data from disparate sources (e.g., linking a P&ID tag to its 3D model location, purchase order, installation IWP, and pre-commissioning loop test).

3.3.4: Functional Modules

The platform must include the following integrated modules:

- **Common Data Environment (CDE) & Engineering Data Hub**
- **Construction Management & AWP**
- **Pre-Commissioning & Mechanical Completion**
- **Commissioning & Startup**
- **Project Controls & Analytics (Cost & Schedule)**
- **Quality & Punch List Management**
- **Materials Management & Traceability**
- **Safety & Permit-to-Work (PTW)**

3.3.5: Detailed Functional Requirements

A. Module 1: Common Data Environment (CDE) & Engineering Data Hub

- **Federated Model Viewer:** Web-based viewer for 3D models (Navisworks, Revit, Plant 3D) with clash detection status and design queries.
- **Document Management:** Version-controlled storage for all engineering deliverables (drawings, specs, datasheets) with automated transmittal and approval workflows.
- **Intelligent P&ID Management:** Digitize P&IDs, enabling users to click on a tag to see its status (e.g., purchased, installed, tested), linked documents, and real-time data.
- **Master Data Registry:** A single registry for the Master Equipment List (MEL), Tag registers, and System/Sub-System breakdowns.

B. Module 2: Construction Management & AWP

- **Digital Work Packaging:** Capability to create, manage, and track Engineering Work Packages (EWP), Construction Work Packages (CWP), and Installation Work Packages (IWP). Each IWP must bundle all required data: drawings, JSA, materials, tools, and procedures.
- **Constraint Management:** A digital Kanban board to track and manage constraints (Design, Materials, Access, Permits) with clear ownership and escalation paths.
- **Progress Measurement:** Automated progress tracking based on IWP completion. Integration with AI-powered field verification (see 4.6).
- **Timesheets & Productivity:** Digital time-capture linked to IWPs for productivity analysis (e.g., earned hours vs. actual hours).

C. Module 3: Pre-Commissioning & Mechanical Completion

This module must digitally enforce the contractual "Instructions to Contractor for Pre-Commissioning."

- **Digital Pre-Commissioning Plan:** A template and workflow for contractors to develop, submit, and execute their Pre-Commissioning and Mechanical Completion Plan within the platform, including:
 - Integrated Pre-Commissioning Schedule
 - System Turnover Sequence definition
 - Vendor Representative schedule management
- **Mechanical Completion (MC) Package Assembly:** Automated compilation of MC Dossiers. The system shall not allow handover until all required documents (certifications, punch list clearance, test reports) are digitally linked and approved.
- **Digital Check Sheets & Procedures:** Execution of all pre-commissioning procedures (flushing, cleaning, calibration, loop testing) on mobile devices. Data auto-populates system records (e.g., Loop Folders, Motor Run Reports).
- **Master Punch List Database:** A single project-wide database for all punch lists. The system shall generate aging reports and link punch items to specific systems, blocking MC until cleared.

D. Module 4: Commissioning & Startup

- **Commissioning Procedure Management:** Digital management of commissioning procedures, including operator training checklists and startup sequences.
- **System Readiness Dashboard:** A dashboard showing the readiness status of each system for startup, integrating data from construction, pre-commissioning, and commissioning.

- **Operational Data Handover:** Secure handover of all asset data (warranties, manuals, test records) to the Operations/Maintenance system.

E. Module 5: Project Controls & Analytics

- **Schedule Integration & Health:** Live integration with Primavera P6. Automated DCMA 14-point schedule health checks and alerts.
- **Quantitative Schedule Risk Analysis (QSRA):** Built-in Monte Carlo simulation engine to calculate P-80/P-90 dates and identify critical path risks.
- **Earned Value Management (EVM):** Automated calculation of SPI, CPI, EAC, and ETC, with trends and forecasts.
- **AI-Powered Predictive Analytics:**
 - Predict design flaws and construction rework based on historical data and model analysis.
 - Forecast potential schedule delays and cost overruns.
 - Recommend optimal construction sequences and resource allocation.

F. Module 6: AI & Digital Field Verification

- **3D Scan to BIM Comparison:** Automated comparison of laser scan/LiDAR point clouds against the design model to verify as-built status and quantify progress.
- **Computer Vision Progress Tracking:** AI analysis of site photos and video to automatically track bulk material placement and equipment installation.
- **Automated Quantity Verification:** Reconciliation of contractor-claimed quantities with AI-verified data from scans and imagery.

G. Module 7: Materials Management & Traceability

- **Digital Warehouse & Logistics:** Track materials from PO to site receipt to final installation using RFID/QR codes.
- **Preservation Management:** Automated scheduling and alerts for equipment preservation routines.
- **Certification Linkage:** Automatic linking of Material Test Certificates (MTCs) to specific tags and IWPs.

H. Module 8: Safety & Permit-to-Work (PTW)

- **Digital PTW System:** Integrated PTW workflow that checks for isolations (LOTO), linked JWAs, and crew qualifications before permit issuance.

- **Real-Time Personnel & Asset Tracking:** Monitor the location of personnel and vehicles in hazardous areas.
- **Incident & Alert Management:** Digital reporting and analysis of safety incidents and near-misses.

3.3.6: Discipline-Specific Pre-Commissioning & Commissioning Checklists

The platform must provide digital checklists and forms for the following disciplines, customizable per project specification:

A. Mechanical (Static & Rotating)

- Hydrostatic Test Packs (Pressure, Hold Time, Medium)
- Equipment Alignment Checks
- Lubrication & Flushing Records
- Pump/Compressor Solo Run Tests (Vibration, Amps, Temperatures)
- Relief Valve Set Pressure Certifications

B. Piping

- Line Cleanliness (Flushing, Air Blowing, Chemical Cleaning) Records
- Torque Records for Critical Bolting
- Hot Bolting Procedures

C. Electrical

- Transformer Turns Ratio & Insulation Resistance Tests
- Cable Meggering & Sheath Test Reports
- Cable Termination Reports and tests records.
- Motor Solo run pump test records
- Relay Setting & Functional Tests
- HV/MV/LV Switchgear Functional Tests
- Records for Fat and SAT for all the equipment's as applicable

D. Instrumentation & Control Systems

- **Loop Checks:** Functional tests from field sensor to DCS/PLC and back to final element. Must record range, alarm setpoints, and response time.

- **Calibration Certificates:** Link to each instrument tag.
- **Control Logic & Interlock Test Reports**
- **SIL Verification Compliance Records**
- **FAT & SAT Reports for DCS/PLC/SIS**

E. Civil & Structural

- Concrete Pour & Test Cube Records
- Structural Steel Bolting/Torque Records
- Foundation Settlement Surveys

F. Process Automation (As per Contractual Requirements)

- **Design Review Tracking:** PDR and CDR workflow with document submission and approval.
- **FAT/SAT Execution:** Digital procedures and reporting for Factory and Site Acceptance Tests, including I/O testing.
- **Functional Loop Testing:** Mandatory digital loop sheets capturing data from sensor to final control element.

3.3.7: Deliverables

The vendor shall deliver the following:

1. **D1:** Fully configured and deployed Integrated Digital Project Delivery Platform with all modules.
2. **D2:** Complete integration with existing Maaden systems (ERP, P6, Document Management).
3. **D3:** A comprehensive Data Migration and Setup Plan.
4. **D4:** A full suite of User Training Materials and a certified Train-the-Trainer program.
5. **D5:** System Administration and Configuration Runbooks.
6. **D6:** A detailed Security & Compliance Dossier, confirming adherence to KSA regulations (NCA, PDPL, CST).
7. **D7:** A Scale-Out Plan for deploying the platform to future Maaden projects.

3.3.8: Key Performance Indicators (KPIs) & Acceptance Criteria

KPIs (To be measured during Pilot Phase):

- **KPI 1:** ≥30% reduction in time from Construction Complete to Mechanical Completion sign-off.
- **KPI 2:** ≥95% of all punch items are logged and assigned an owner within 24 hours of identification.
- **KPI 3:** ≥50% reduction in time spent by engineers compiling handover dossiers.

- **KPI 4:** 100% of pre-commissioning test records (e.g., loop sheets) are digital and directly linked to asset tags.
- **KPI 5:** $\geq 15\%$ improvement in Plan Percent Complete (PPC) for pilot work areas.
- **KPI 6:** The platform provides a forecasted delay alert for a critical path activity at least 4 weeks in advance.

3.3.9: Acceptance Criteria:

- All deliverables (D1-D7) are fully implemented, configured, and operational.
- The platform successfully processes a full, digital handover of one pilot system from Construction through to Ready for Startup (RFSU).
- All KPIs are met or exceeded for the pilot scope, or a justified improvement plan is accepted by Maaden.
- Positive user acceptance feedback is received from key stakeholder groups (Project Controls, Construction, Pre-Commissioning, and Contractor personnel).

3.3.10: Implementation, Timeline & Support

- **Phase 1 (Weeks 1-4):** Requirement Finalization, Environment Setup, Core Configuration.
- **Phase 2 (Weeks 5-12):** Pilot Module Deployment (AWP, Punch List, Pre-Commissioning) on a selected project area.
- **Phase 3 (Weeks 13-16):** Pilot Validation, KPI Measurement, User Training, and Feedback Incorporation.
- **Phase 4 (Weeks 17+):** Full-Scale Rollout across the entire ARGP project.
- **Support:** The vendor must provide 24/7 support with a defined SLA and a dedicated account manager.

3.4 Module 4: Quality Management & Assurance

3.4.1: Objectives

MAADEN seeks to digitize and automate Quality Management processes across engineering, procurement, construction, pre commissioning, and handover. The solution must reduce cycle time, enhance traceability and accountability, provide real time analytics, and ensure compliance with ISO 9001 and owner/operator standards flexibility of integrating with existing owner systems.

3.4.2: General

Define the architecture for a centralized digital quality platform to manage reporting, documentation, audits, and task tracking, ensuring alignment with Maaden's enterprise architecture standards.

Implement a secure, cloud-based document management system with user-specific access control, version history, encryption protocols, and compliance with Saudi cybersecurity standards (NCA).

Enable real-time reporting, automated notifications, and escalation workflows for timely decision-making and issue resolution within the quality function.

Ensure seamless integration with existing enterprise systems such as Oracle, Power BI, Aconex, and Microsoft Outlook to promote data continuity and reduce duplication.

Design the platform to be scalable and modular, supporting API integrations with other corporate systems, and include data backup and disaster recovery capabilities.

Develop a structured deployment and change management plan, including end-user training, onboarding documentation, and ongoing support to ensure successful adoption across departments and sites.

Digitize quality tools such as checklists, forms, and inspection records, accessible across mobile and desktop devices to support field-level data collection.

Automate assignment and tracking of corrective and preventive actions (CAPA), with defined deadlines, ownership, and follow-up notifications.

Include interactive dashboards for real-time visualization of quality KPIs, nonconformities, and audit outcomes to support data-driven decision-making.

Incorporate a full audit trail to monitor and log all user interactions for compliance, accountability, and traceability.

Enable role-based workflows for approvals, escalations, task delegation, and authority matrix alignment.

Support multilingual interface and user customization, tailored to departmental functions, levels of access, and user preferences.

Ensure compliance with ISO 9001 and other relevant national/international quality standards through pre-built templates, standard operating procedures, and version-controlled forms.

Allow offline data entry functionality, with automatic synchronization once reconnected to the network to support remote site operations.

Ensure the system is mobile-responsive and cross-platform compatible, enabling access from PCs, tablets, and smartphones.

Establish a structured feedback mechanism within the system to collect user suggestions and drive continuous improvement initiatives.

Incorporate advanced analytics and AI capabilities for predictive quality insights, anomaly detection, and process optimization recommendations where feasible.

Define system ownership and governance roles between Quality, IT, and relevant stakeholders to ensure platform sustainability, compliance, and alignment with digital strategy.

Implement a phased rollout approach, starting with pilot departments or sites, allowing for testing, refinement, and scaling across Maaden business units.

Phase 1 — Baseline Modules

Request for Inspection (RFI/MIR): schedule, call-out windows, hold/witness points, e-signatures, rescheduling with reasons, attachments (Photos/ITR/COA), TPI call-outs.

Nonconformance (NCR) & Quality Observation Report (QOR): defect/CAUSE, CAR/PAR workflow, containment → corrective → verification, escalation from QOR to NCR, cost tracking.

Preservation Management: tag/equipment register, OEM preservation tasks and intervals, checklists with photo/geo stamp, overdue alerts, storage/condition logs, linkage to NCR.

ITPs & ITRs: discipline ITPs withhold/witness, mobile ITR capture (offline), acceptance criteria, rejection/retake logic.

Project Quality Index (PQI): controlled distribution, revision control, read & acknowledge tracking, exception logs.

Dashboards & KPIs: NCR cycle time/aging, QOR heatmap, ITP compliance, punch closure, inspector productivity.

Vendor Inspection & Documentation: vendor ITPs, inspection notifications, visit reports, release notes, MDR.

Walkdowns & Punch Listing: A/B/C category, owner/due date, subsystem/system, evidence, integration to completions.

Phase 2 — Next Modules (to implement after stabilization)

AI Capabilities Handover Packages / Dossiers: electronic turnover packs (system/subsystem), auto-compile ITR/NCR/QOR/MDR, e-signatures, owner templates.

NDT Module: UT/RT/MT/PT report capture, technician qualifications, acceptance criteria mapping, traceability to tags/welds/lines.

Piping Test Packages: hydro/pneumatic test packs, boundaries & iso split, pressure logs, gauge calibration linkage, leak/retake workflow, reinstatement.

- Audit Register
- Delivery approach

As-Is Assessment: map PQI, ITP/ITR, NCR, QOR, RFI/ MIR, punch, preservation, vendor docs; capture pain points and compliance gaps.

Requirements & To-Be Design: business requirements, forms/fields, roles, RACI, workflow states, SLAs, evidence rules, retention; Arabic/English content.

Process & Form Design (non-IT): finalize forms, validations, acceptance criteria, escalation paths; no technical architecture in scope.

Interface Requirements (business-only): catalogue of business data exchanges (objects/fields/frequency) to hand to IT.

Pilot Configuration: configure Phase-1 modules (RFI/ MIR, NCR & QOR, Preservation, ITP/ITR, PQI, Dashboards, Vendor Inspection & Docs, Walkdowns & Punch) in a controlled area.

Training: quick-start guides and train-the-trainer for pilot users; adoption plan.

3.4.3: Functional Requirements

- The system shall allow users to create, assign, and track quality-related tasks with defined due dates, escalation rules, and responsible owners.
- The system shall generate automated reports on a daily, weekly, and monthly basis, aligned with predefined quality KPIs.
- The system shall support centralized and secure storage of quality documents, with version control, access logs, and audit history.
- The system allows real-time data entry through both web and mobile interfaces, ensuring field and office users can input data efficiently.
- The system shall issue automatic notifications and reminders for overdue tasks and pending approvals based on configurable rules.
- The system should provide interactive dashboards for visualizing nonconformities, audit results, CAPA status, and other performance metrics in real time.
- The system shall integrate with enterprise systems such as Oracle (or equivalent ERP) to retrieve project, material, and inspection data seamlessly.
- The system shall maintain a complete audit trail of all user actions, including task updates, approvals, document changes, and system events for full traceability and accountability.
- The system shall support role-based access control (RBAC) to restrict access and actions based on user roles, responsibilities, and organizational hierarchy.
- The system shall allow users to generate and export reports in multiple formats, including Excel, PDF, and CSV.
- The system shall support form-based digital input for quality checklists, inspection forms, evaluations, and assessments.

- The system shall enable offline data entry, with automatic synchronization when the device reconnects to the network, to support remote and low-connectivity environments.
- The system shall be designed with modular architecture to enable future expansion, feature enhancements, and system upgrades without major redesign.
- The system should include multi-language support and a customizable user interface, tailored to departmental needs and user preferences.
- The system shall support automated backup, data recovery, and high availability to ensure business continuity and prevent data loss.
- The system shall enforce compliance with Maaden's cybersecurity policies and national regulatory standards (e.g., NCA) to ensure secure handling of sensitive data.
- The system shall provide configurable workflows for approvals, reviews, task escalations, and quality gate processes across departments.

3.4.4: Technical Requirements

Application Architecture:

- The solution design should be capitalized on the existing enterprise solutions first. If it's not available as a native feature, then extensions to be used. If the volume of work is bigger than the extension capability, a new product to be selected considering the integration and scalability capabilities is available.
- Applications will be modular and independent ("atomic") in nature. They will access common data, use common services, and have only inherently essential dependence on other applications (e.g., for provision of up-to-date data).
- Components and systems will be reusable. All IT solutions will be assembled from an agreed set of existing, sharable components, whenever possible
- All new applications shall use Maaden standard, defined methodology and products, including application development languages, compilers and tools and will utilize a consistent application development methodology, when developing applications internally or through an outsourcer.
- Applications are independent of specific technology choices and therefore can operate on a variety of technology platforms.
- Users will be provided with a consistent and easy to use interface to access all authorized applications, data and services at any appropriate location and Applications will meet user defined criteria for ease of learning, use, and support.
- Outsourced activities must not be exceptions to the enterprise architecture simply because they are provided by a third party.
- All decisions pertaining to Oracle ERP Fusion Cloud (Enterprise Resource Planning) Enhancements must consider the Cloud computing approach leveraging the SaaS (Software as a service) and PaaS (Platform as a Service) services offered by Cloud product vendors. Cloud computing represents a significant shift in the way that IT resources are managed, operated, and consumed. This change exposes several benefits, promoting greater IT efficiency and agility.
- Reporting & Analytics processes to be optimized and aligned with business perspective.
- Authentication Shall be done via Microsoft ADFS used by Maaden.
- The solution design shall use Maaden approved Database Technologies (Microsoft, Oracle) otherwise the solution provider shall provide the necessary information for the need of the new proposed Database Solution to be adopted by IT technology roadmap.

- The Solution Design shall pass performance testing during the testing phase of the implementation project.
- The solution provider shall provide all the necessary documentation having:
 - IT Admin Manual
 - Super User Manual
 - End User Manual
 - Implementation Manual (installation steps with screenshots)
 - Technical Requirements Manual
 - Functional Requirements Manual
- The solution shall have the capability of Administration module – allowing Superuser/ power user to operate and manage the business-related configuration
- The solution design shall have the Integration Capability to be integrated with Maaden Middleware.
- The solution design shall have a proper Useability.
- The solution design shall support the Mobility features by having the ability to be extended to Maaden low-code platform or having a native mobile application registered on the marketplace.
- The solution development shall follow the DevSecOps framework utilizing Maaden ALM tool.

Data Integration & Extraction

- Maaden assumes the ability to integrate, and extraction of data can be at any point of time without technical restrictions applied on the solution.
- Any technical restriction must be communicated and acknowledged by Maaden data team. Provide insights on how the data design may impacts application performance if there would be a scenario.
- List all types of connectors that can be used by Maaden team to integrate & extract data, APIs are a must to be provided with full exposure on the data layer and without rate limitations.
- Maaden assumes the ability to integrate with solutions such as Informatica, Denodo, and its Oracle Data Warehouse, Power BI are feasible, it must be communicated and acknowledged by Maaden data team in case any restrictions are there.
- APIs are expected to follow modern authentication/ authorization best practices and RESTful architecture.
- Any performance degradation of the Application due to data integration/ extraction are to be communicated with Maaden team, along with best practices to ensure no hiccups in production environment.
- Data Lifecycle Management The solution provider shall share the policies for data archiving, purging, and backup and ensure they are aligned with Maaden policies. Include details about data retention periods and how data archiving is handled.
- Development Maaden assume a non-production environment with scrambled data for testing/ development purposes exists
- Data Dictionary/ Documentation The documentation of the data layer of the solution must be well detailed and having well written examples on data querying and manipulation (if applicable)
- Data tables shall have clear definitions along with each attribute usage/ purpose Compliance and Legal Requirements
- Maaden assumes the adherence to Saudi Personal Data Protection Law (PDPL) by the solution provider and share necessary evidence in form of reports regularly (to be defined with focal point later) to demonstrate the adherence.
- Any noticed incompliance with the clauses of PDPL are to be shared with Maaden immediately, focal point of Maaden must acknowledge it.

IT Security Requirements

- Solution must support open identity federation standards (SAML, WS-Federation, OAuth 2.0, etc.) as a means of authenticating/authorizing Maaden users for its offered products and services. More specifically, the proposed Solution should be able to integrate with Microsoft Azure AD for user single sign-on (SSO) authentication.
- Solution must be able to implement encryption of data in transit for Maaden's data using industry best practice and standard for encryption algorithm and keys management. Transport layer security implemented at the web access layer using HTTPS/TLS must meet or exceed the industry's security best practice standards (e.g., minimum TLS v1.2).
- The Solution must support role-based access control to manage access for users by assigning appropriate roles to do the job.
- Respondent must conduct application vulnerability assessments and penetration tests in the Solution by qualified individuals (e.g., internal and/or external 3rd parties) and address security vulnerabilities in a timely manner.
- Ability to generate API access keys.
- The bidder shall provide the solution which fully support matured authentication and authorization capabilities and services such as but not limited to ADFS, SSO, MFA, segregation of duties, least privilege, role-based access, etc.
- The bidder shall provide the solution which encrypts data in motion and data at rest with AES-256 or higher encryption standard.
- The bidder shall provide the solution which has capability to support data classification.
- The bidder shall provide the solution which be hosted within KSA.
- The bidder shall provide the solution applies high availability (redundancy) and is hosted in different regions (disaster recovery).
- The bidder shall provide the solution uses security controls such as but not limited to firewalls, WAF, API protection, IPS/IDS, antivirus, EDR, etc.
- The bidder shall provide the solution obtains hosting provider certification such as but not limited to ISO, SOC 2.
- The bidder shall provide the solution on which penetration testing, vulnerability assessment, risk assessment, patch management, and backup are conducted periodically.
- The bidder shall provide the solution has matured incident response plan.
- The bidder shall provide the solution integrates with the central logging system, Maaden SIEM to capture all relevant applications and system events.
- The bidder shall provide the solution is monitored by 24/7.
- The bidder shall be responsible to comply with Maaden cybersecurity requirements.
- The bidder shall provide Maaden with all required documents such as but not limited to incident response plan and assessments plans & reports, and backup strategy & retention.
- It is the responsibility of the provider to seek and validate Maaden understanding of the requirements.
- Maaden understanding of the requirements supersedes any other understanding or interpretation.

3.4.5: Project Deliverables and Documentation

The selected Service Provider must successfully deliver the project and comply with following requirements:

1. Maaden Project Delivery Framework and related requirements for each stage gate.
2. Project plan and content, functional, technical and other guide materials considering business flows and test scenarios to be available as soft and approved copies on MS EPM System.
3. High quality training to end users, super users, and knowledge transfer to ICT Support in addition to materials as per Maaden processes.
4. Built product extensions configuration should be documented in version control system upon knowledge transfer completion with ICT support in addition to MS EPM system.
5. 1 Month hyper-care after project Go live to be provided by the implementer with no additional charges to fix reported issues considering agreed project resource plan and service level agreement.
6. Hyper-care extension for the project is only upon Maaden request.

3.4.6: Implementation Duration and Location

As per Maaden requirements, this project should complete within XX months, taking in consideration the below factors:

- Maaden project team is fully committed during accountable milestones of implementation as delineated in the project plan till project sign-off and Go-live.
- Implementer should define prerequisites to be prepared by Maaden side. Communicated items should include business and technical requirements to help the implementer to achieve desired deliverables once awarded and mobilized.
- The implementation team will be stationed in Ras Al-Khair (RAK). However, based on project needs, it might be required for selected team members to be stationed in various Maaden Affiliates (e.g. Corporate Headquarters in Riyadh and other areas).

3.4.7: Project Implementation

In the following sub-sections, Implementer is expected to provide all required information regarding the implementation of this project.

- **Confirmation of Functional and Technical Requirements:** The Systems Integrator is expected to go through SOW document and highlight how to implement each listed function in the required applications.
- **Maaden Information Security Requirements. On transit between Cloud and Maaden Applications Domain:** Data should be SSL-encrypted. Minimum 128-bit encryption. The protocol used for exchange of data between cloud and onsite applications should be HTTPS only.
- **On transit between Maaden and Cloud Applications:** Data should be SSL-encrypted. Minimum 128-bit encryption.
- The protocol used for exchange of data between onsite applications and cloud should be HTTPS only.
- **Implementation Approach:** Implementer is expected to describe the proposed implementation and cite a project where the methodology has been recently applied. Implementer should address the solution deliverables and explain alternatives that are most feasible, from Implementer's perspective considering best practices that have been implemented successfully.
- Implementer is expected to describe the nature of proposed solution with details on the following:
- **Product extensions** development, if necessary, considering justified requirements.

- **System functionalities testing** (unit, system, regression, user acceptance testing and rest of the tasks).
- Integration development and testing.
- **Data conversion programming and testing:** The Implementer is requested to discuss preferred conversion approach, including the methodology for determining data elements to be converted, amount of history to be converted, accountability for the data conversion between (e.g., the Implementer or Maaden). This approach should reference data conversion methodology for similar clients.
- System and user acceptance testing considering test scenarios.
- Orientation and user training: Describe the approach for user training, including information on training allowances bundled in cost, how training costs are determined, limits on number of employees covered by allowance and location of training (e.g. on site or online).
- Documentation as per Maaden process.
- The Implementer may add additional points that are not covered and will support the proposed approach technically.
- **Implementation Project Schedule:** Implementer should submit a plan that demonstrates approved methodology in implementing desired solution and outline specific activities that must be accomplished to achieve an orderly conversion and implementation. The implementation task plan should identify an estimated timeline for the implementation of all proposed products, items in scope and level of efforts (in hours or days) required of the Implementer resources and Maaden project team. The implementation task plan should also identify specific milestones and the acceptance criteria for each milestone on MS EPM System.
- **Project Implementation Team:** Implementer is expected to describe the proposed project team for the implementation and the areas of the implementation that they will be involved in. As a starting point the following roles are seen to be required considering numbers and additional roles as required:
 - Executive Sponsor.
 - Systems Integrator Project Manager.
 - Senior Functional Consultant\s.
 - Senior Technical Consultant\s.
 - Senior Database Administrator.
 - Documents Controller.
- **When proposing resources, the following requirements are imperative:**

Project manager is expected to be very senior resource and carried such implementation before.

Deploy highly experienced functional onsite and technical consultants for implementation.

Resources should be experienced with indicated products and integration technologies on cloud or on-prem where applicable for the implementation and development.
- Detailed resumes that demonstrate the background and accomplishments in similar projects must be included. Maaden will interview all proposed resources and select the most suitable one for this project.
- Maaden expects qualified resources to be engaged in this project. In all cases, Maaden reserves its right to require resource replacement without affecting the project timelines if performance of any resource is not satisfactory. Furthermore, Maaden will need to approve any resource transfer from the project. Any agreed transfer cannot proceed until a suitable replacement is onboard.
- **Project Management and Problems Resolutions:**

The Implementer should describe each resource role, project manager, functional and technical team in addition to assigned executive sponsor who's going to present for escalations and steering committee events.

Project manager is expected to use MS EPM system, follow Maaden process and deliver the following but not limited to:

- Progress monitoring and status reports on regular basis.
- Project schedule review and constrains managements.
- Manage weekly meeting and project related events.
- Issue & Risk management.
- Key success factors management.

The Implementer should provide problem resolution and escalation process that involve the assigned sponsor and highest level in the Implementer's firm hierarchy.

3.4.8: Project Initiation Requirements:

The Implementer should describe the involvement required by Maaden in terms of IT and other business resources for successful implementation:

- What key roles will need to be filled by Maaden staff?
- What is the typical amount of involvement from Maaden team members on a per week basis?
- Which level of expertise and authority power required?
- What is logistics required to be prepared?
- What is the software\hardware required?

All requirements and preparations clearly to be indicated and confirmed prior to the implementation kick-off to ensure smooth project delivery.

3.4.9: System Integrator to fill in as a response:

Assumptions & dependencies:

Project Staffing Plan: Describe if the Services are based on time & material basis or fixed-cost, describe the projected/agreed staffing plan, duration and all resources required

MAADEN responsibilities and tasks: Describe if there are specific tasks and responsibilities that MAADEN is required to perform

Service Performance and Delivery Methodology Describe the project execution methodology, if applicable

Description of Deliverables: Be specific and only list those tangible Deliverables that will be provided to the MAADEN. Everything else should only be described in the section on Services.

Third-Party Components: Should describe all third-party items and materials to be included or provided as part of any Deliverables.

Term of the Project: Project Schedule; Milestones, Include dates for start and completion of the each phase of the Services, and for delivery of each of the Deliverables.

Contingencies and risk mitigation plan: Review and Testing; Acceptance:

Describe the review and testing process and indicate whether the Review and Testing Period will be longer or shorter than the default period of 30 days set forth in the Agreement. Where possible, specify the acceptance testing that MAADEN will perform on each Deliverable.

Compensation: Indicate the fees and compensation payable and whether the project is a time and materials project or a fixed-fee engagement. Also indicate when System Integrator is entitled to invoice MAADEN for fees and expenses (e.g., monthly, or on the achievement of Milestones, Acceptance of Deliverables, etc.). Indicate whether invoicing for fees and expenses will occur other than on a monthly basis. If the fees are based on time and material rates, specify if per diem and round trip air travel reimbursement apply for assignments of less than 90 days

Assumptions: The system will be hosted on a secure cloud platform with compliance to Maaden and NCA cybersecurity requirements.

End users and key stakeholders will be available to participate in requirement validation, and training sessions.

Maaden will assign a dedicated project owner from the quality function to coordinate decisions and approvals.

Any additional languages or regions to be supported will be specified prior to the final UI design phase.

Any enhancements requested after go-live will follow Maaden's formal change request process.

3.4.10: Core Capabilities

- **Digital Quality Control Plans (ITPs):** Execution of digital Inspection & Test Plans with enforced hold points.
- **Non-Conformance Report (NCR) Management:** End-to-end digital workflow for NCRs, CAPA, and root cause analysis.
- **Supplier & Contractor Quality Performance:** Digital scorecards based on NCR rates, CAPA closure, and adherence.
- **Material & Welding Traceability:** Full traceability from Material Test Certificates (MTCs) and weld maps to installed components via QR/RFID.
- **AI-Predictive Analytics:** Forecast quality failure points based on design, vendor history, and site data.

3.4.11: Key Deliverables & KPIs

- **Deliverables:** Fully configured digital QMS, automated Quality Dossier compilation, supplier performance dashboards.
- **KPIs:** 100% of NCRs logged within 24 hours; ≥40% reduction in NCR closure time; 100% digital traceability for critical welds.

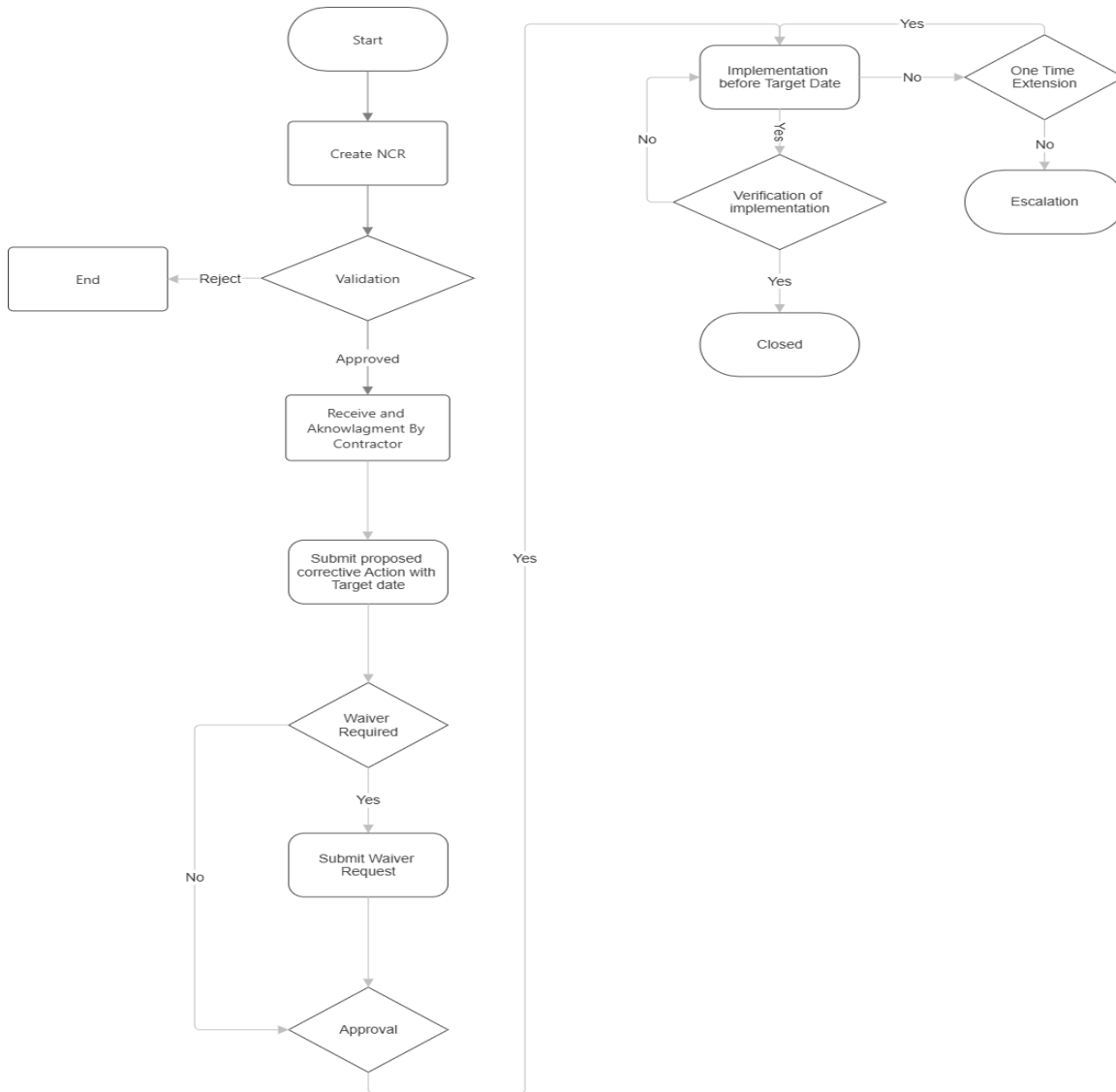
3.4.12: Process Flow

Non-Conformance Report (NCR)

	Client										Contractor and Sub-Contractor		
	Admin	Project Manager	Project Director	Quality Director	Quality Manager	Quality Lead	Inspector	TPI	DC	Viewer	Quality Manager	Inspector	DC
Create NCR						R	R						
Validation						V/ S							
Receive and Acknowledgement											A	I	I
Submit proposed Corrective Actions												R	
Submit Waiver Request											R		
Approve Corrective Actions						I	A						
Implementation of Corrective Actions											R		
One-Time Extension						A					R		
Verification of Implementation						A	R						
Escalation						R							
Close NCR						A/ S	V/ S						

- R – Responsible.
- A – Accountable (Approver)
- C – Consulted,
- I – Informed

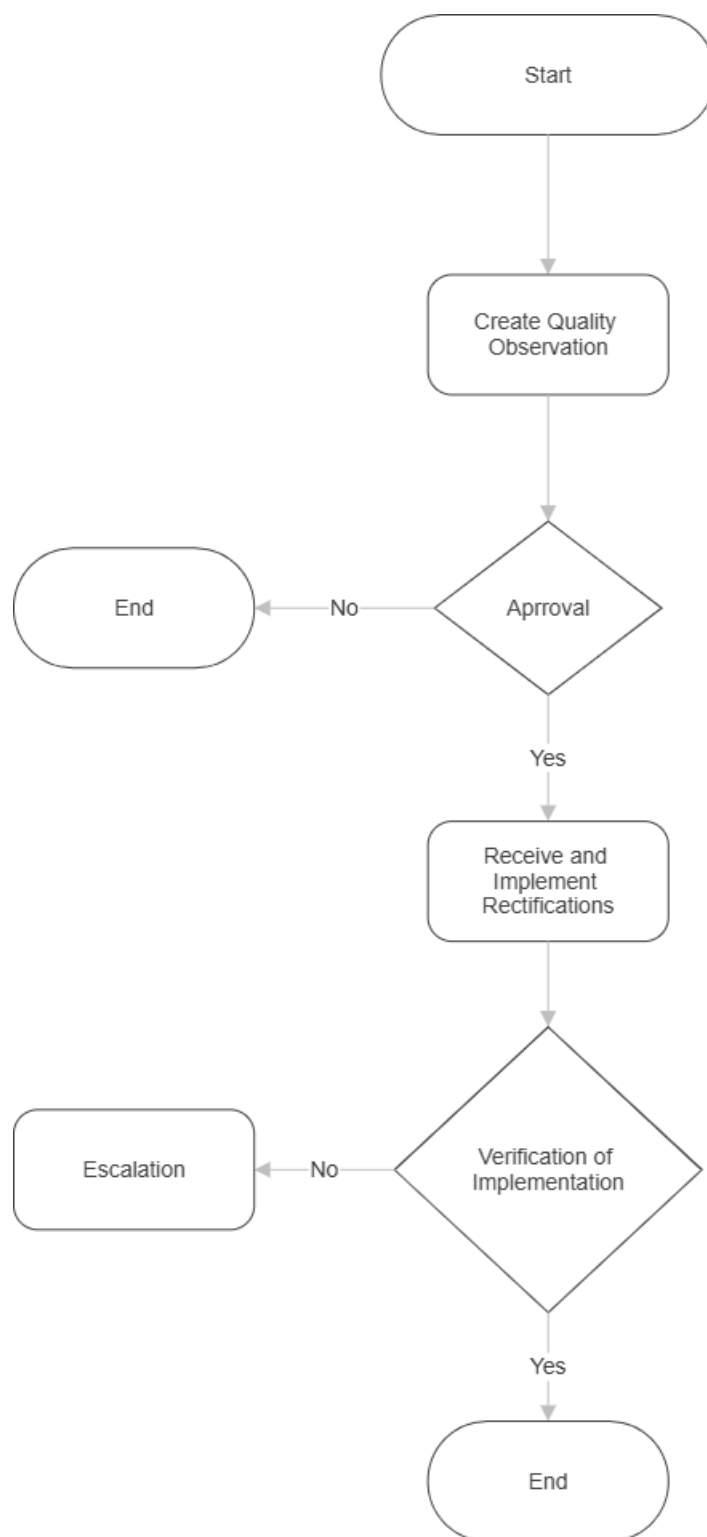
- V – Verifier
- S – Signatory



Quality Observation Report (QOR)

	Client										Contractor and Sub-Contractor		
	Admin	Project Manager	Project Director	Quality Director	Quality Manager	Quality Lead	Inspector	TPI	DC	Viewer	Quality Manager	Inspector	DC
Create Quality Observation						R	R	R					
Approving the Quality Observation					I	V/ S							
Receive and Implement Rectification											A	R	I
Verification of Implementation						A	R						
Escalation				I	I	R					A	R	
Close Observation				I	I	A/ S	V/ S						

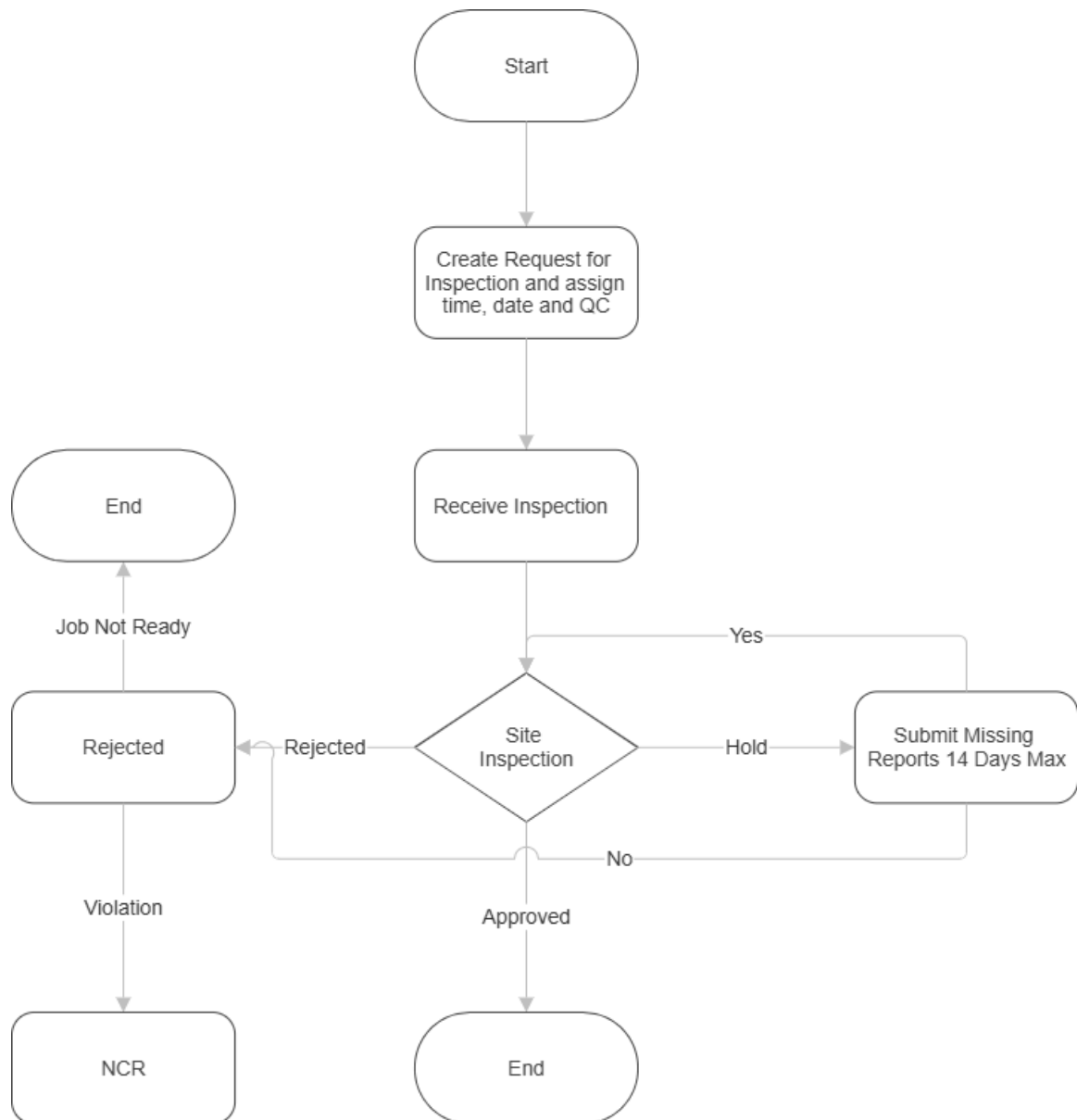
- R – Responsible.
- A – Accountable (Approver)
- C – Consulted,
- I – Informed
- V – Verifier
- S – Signatory



Request for Inspection (RFI)

	Client										Contractor and Sub-Contractor		
	Admin	Project Manager	Project Director	Quality Director	Quality Manager	Quality Lead	Inspector	TPI	DC	Viewer	Quality Manager	Inspector	DC
Create Request and assign QC									I		V	R/S	I
Receive Inspection						I	I	I	R		R	R	R
Site Inspection Report							R/V	R/V				R	
Submit Missing Reports						A	V	V				R	
Rais NCR						V/S	R	R					
Close the RFI						I	A/S	A/S				R/S	R

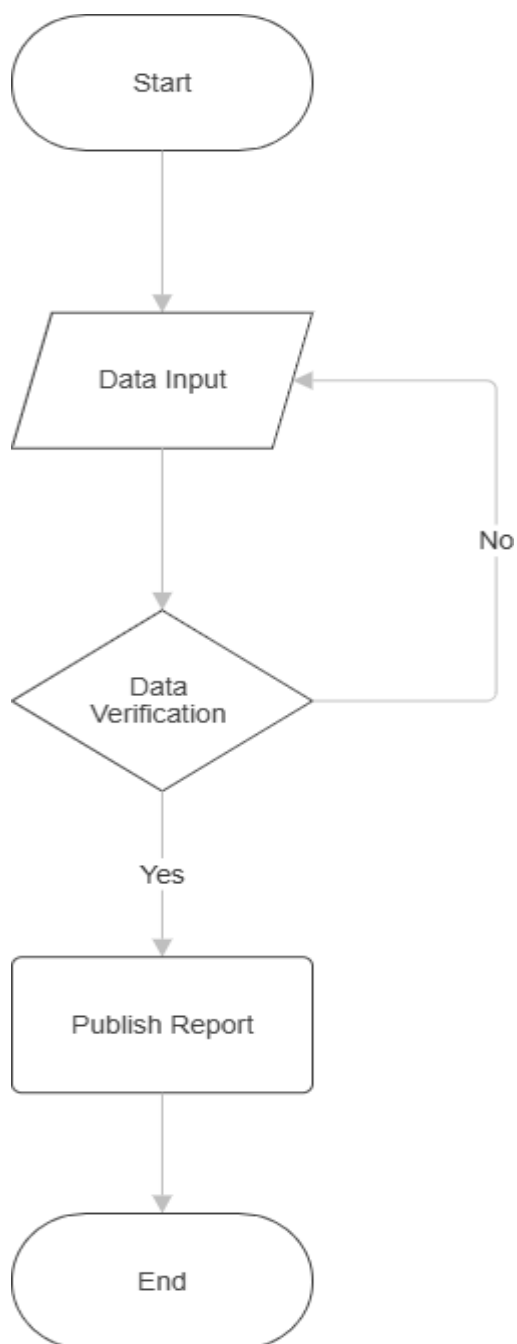
- R – Responsible.
- A – Accountable (Approver)
- C – Consulted,
- I – Informed
- V – Verifier
- S – Signatory



Project Quality Performance (PQPI)

	Client										Contractor and Sub-Contractor		
	Admin	Project Manager	Project Director	Quality Director	Quality Manager	Quality Lead	Inspector	TPI	DC	Viewer	Quality Manager	Inspector	DC
Data Input				I	I	R					R		
Data Verification			I	I	C/S	R/S							
Publish Report		I	I	I	I	I							

- R – Responsible.
- A – Accountable (Approver)
- C – Consulted,
- I – Informed
- V – Verifier
- S – Signatory



3.5: Module 5: Procurement & Supply Chain Management

3.5.1 Objectives

Create a transparent, synchronized, and intelligent procurement ecosystem to prevent delays and optimize the supply chain.

3.5.2 Core Capabilities

- **End-to-End Procurement Workflow:** Digital management from Requisition to PO, Expediting, and Payment.
- **Supplier Management & Risk Analytics:** Centralized supplier database with AI-driven risk scoring.
- **Expediting & Logistics Synchronization:** Real-time dashboards and alerts against "Need-by" dates from the project schedule.
- **Vendor Data Management:** Centralized portal for vendor document submission with AI-compliance checks.
- **Materials Management:** Track materials from receipt to installation using QR/RFID, including preservation management.

3.5.3 Key Deliverables & KPIs

- **Deliverables:** Fully configured procurement module with supplier portal, integrated expediting dashboards.
- **KPIs:** ≥95% of critical POs integrated with schedule; ≥30% reduction in time chasing vendor docs; 4-week early warning for ≥90% of late deliveries.

4. Platform-Wide Requirements

4.1 Technical Architecture & Integration

- **Cloud-Native & Scalable:** SaaS-based, multi-tenant architecture capable of supporting multiple mega-projects.
- **Open API Framework:** Robust, secure APIs for bidirectional integration with all core systems (ERP, P6, CAD/BIM, Document Management).
- **Semantic Data Model:** An ontology-driven model to relate data across all modules (e.g., linking a P&ID tag to its PO, installation IWP, and test record).

4.2 Data Security, Residency & Compliance

- **KSA Data Residency:** All project data must reside within the Kingdom of Saudi Arabia.
- **Security Standards:** Compliance with ISO/IEC 27001, NCA ECC-1:2018, KSA PDPL, and CST Cloud Computing Regulatory Framework.
- **Access Control:** Strict role-based access controls (RBAC), multi-factor authentication (MFA), and immutable audit trails.

4.3 Implementation, Support & Deliverables

- **Phased Implementation:** A detailed plan for Pilot deployment (targeting high-impact areas) followed by full-scale rollout.
- **Support Model:** 24/7 support with defined SLAs and a dedicated account manager.
- **Comprehensive Deliverables:** Include configured platform, integration connectors, training materials, runbooks, security compliance dossier, and a scale-out plan.

5. Vendor Response Requirements

The bidding vendor must address all aspects of this SoW and explicitly demonstrate the following in their proposal:

1. **Platform Capabilities:** Proven experience with a platform capable of delivering the integrated modules described.
2. **AI & Analytics Demonstration:**
 - Conduct a live QSRA on sample data, presenting S-curves and tornado charts.
 - Demonstrate "what-if" optimization and AI-powered predictive alerts.
 - Showcase discipline-specific AI design checks.
3. **Integration & Technical Prowess:**

- Demonstrate robust integration with Primavera P6 and ERP systems.
- Showcase the semantic data model and ontology management capabilities.

4. Compliance & Commercial:

- Provide explicit compliance mappings for NCA ECC-1:2018, PDPL, and CST CCRF.
- Confirm in-Kingdom data hosting options.
- Submit a detailed implementation plan, timeline, resource model, total cost of ownership, and SLA definitions.

5. Explicit Solution for Stated Requirements:

- End-to-end schedule build and large-scale simulation capabilities.
- Constraint-based optimization for critical path compression.
- EAC configuration integrated with schedule risk.
- Digital workflow enforcement for pre-commissioning and handover.

6. Standards & Regulatory References (KSA & International)

This SoW mandates compliance with the following standards and frameworks:

1. ISO 19650 (Building Information Modelling & Common Data Environment)
2. ISO/IEC 27001 (Information Security Management)
3. ISO 9001 (Quality Management Systems)
4. Construction Industry Institute (CII) Advanced Work Packaging Standard
5. Saudi NCA ECC-1:2018 (Essential Cybersecurity Controls)
6. Saudi Personal Data Protection Law (PDPL)
7. CST Cloud Computing Regulatory Framework (CCRF)

7. Vendor Response Requirements (Software Developer)

Software Developer must demonstrate:

1. Proven Philosophy and lineage management capabilities.
2. AI-enabled compliance and optimization tools aligned with Maaden standards.
3. Seamless integration with CAD, BIM, ERP, and construction management systems.
4. Compliance with PDPL, ECC, and CCRF requirements.
5. A pilot execution roadmap demonstrated measurable KPI improvements.
6. Demonstrate robust P6 integration, governance workflows, and lineage preservation.
7. Run a QSRA on anonymized sample data; present S-curves, tornado drivers, and quantified mitigations.

8. Show what-if/optimization capabilities with constraint awareness and impact tracking.
9. Provide EVMS integration, contractor scorecards, and executive dashboards with drill-downs.
10. Submit PDPL/ECC/CCRF compliance mappings; KSA residency options; EULA, update cadence, and support SLAs.
11. Provide training, runbook, and scale-out plan with resource model and timelines.
12. Provide requirements related to Baseline schedule, such as Level 4, Resource loaded (Quantity & Craft wise Manpower) Cost loaded for conducting schedule & Cost Optimization Requirements related to Cost for forecasting EAC
13. Implementation Plan for setting up the Construction Management Software.

END OF SCOPE OF WORK